



Associate in Applied Science

Cybersecurity (A25590CX)

Red Team Career Track (Ethical Hacking & Pen Testing)

Overview

Cybersecurity professionals protect the systems, networks, and data that organizations depend on—and demand for them in the Charlotte region has never been higher. You'll build a strong security foundation in networking, operating systems, Linux, virtualization, protocol analysis, and security concepts, then specialize in a career track.

The Red Team track trains you to think like an attacker—perimeter defense, ethical hacking, and Python-powered exploitation—so you can find and fix vulnerabilities before adversaries do. Coursework aligns with Certified Ethical Hacker (CEH) and PenTest+.

Employment Outlook

Organizations need people who think like attackers. Charlotte-area employers have posted 1,525 Cybersecurity Engineer and 798 Application Security Engineer roles calling for offensive-security skills, often with entry salaries of \$60,000–\$70,000—within a regional market of 17,000+ openings averaging nearly \$125,000.

Your degree can open doors to roles like Penetration Tester, Ethical Hacker, Cybersecurity Engineer, Application Security Engineer, and Vulnerability Analyst.

(Please note: CPCC does not guarantee employment)

Plan of Study

CREDITS

Term I

ENG 111 Writing and Inquiry	3
MAT 143 Quantitative Literacy	3
CIS 110 Introduction to Computers	3
CTI 110 Information Technology Foundations	3
ACA 122 Transfer & Career Success	1
Term Credits	13

Term II

CTS 120 Hardware/Software Support	3
NOS 110 Operating Systems Concepts	3
NOS 120 Linux Single User	3
CTI 120 Network and Security Foundations	3
PHI 240 Introduction to Ethics	3
Term Credits	15

Term III

NET 125 Introduction to Networks	3
NET 126 Switching and Routing	3
SEC 110 Security Concepts	3
CTS 115 Information Systems Business Concepts	3
PSY 150 General Psychology	3
<i>You may have completed a program certificate(s). Confirm eligibility with your academic advisor.</i>	
Term Credits	15

Term IV

CCT 250 Network Vulnerabilities I	3
SEC 151 Introduction to Protocol Analysis	3
CCT 121 Computer Crime Investigation	4
SEC 175 Perimeter Defense	3
SEC 185 Ethical Hacking	3
<i>You may have completed a program certificate(s). Confirm eligibility with your academic advisor.</i>	
Term Credits	16

Term V

ENG 112 Writing and Research in the Disciplines	3
CTI 140 Virtualization Concepts	3
Technical Elective	1
CSC 211 Ethical Hacking With Python I	3
SEC 255 Ethical Hacking Tools	3
<i>You may have completed a program certificate(s). Confirm eligibility with your academic advisor.</i>	
Term Credits	13

Total Credit Hours

72